



Event Relations and Probability Rules

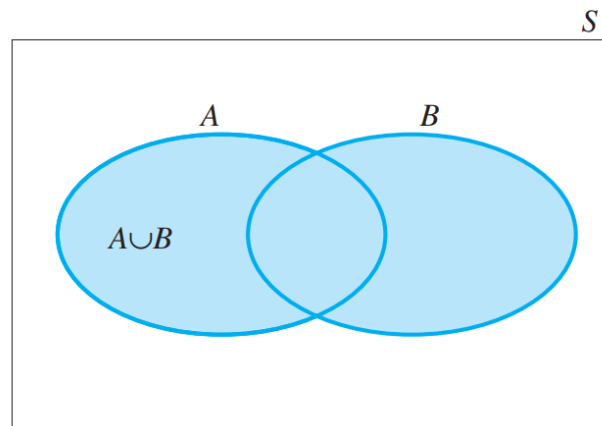


Event Relation

Sometimes the event of interest can be formed as a combination of several other events. Need to consider 3 important event rules (same as set theory)

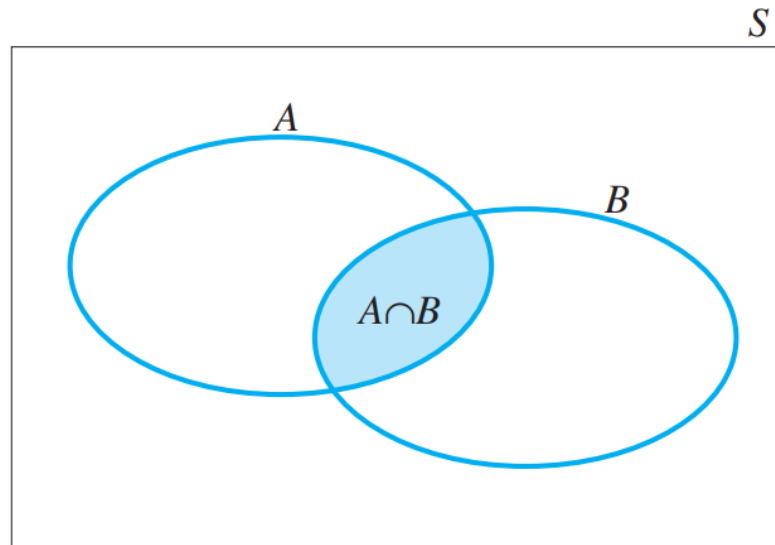
The Union Rule

The union of events A and B , denoted by $A \cup B$, is the event that either A or B or both occur.



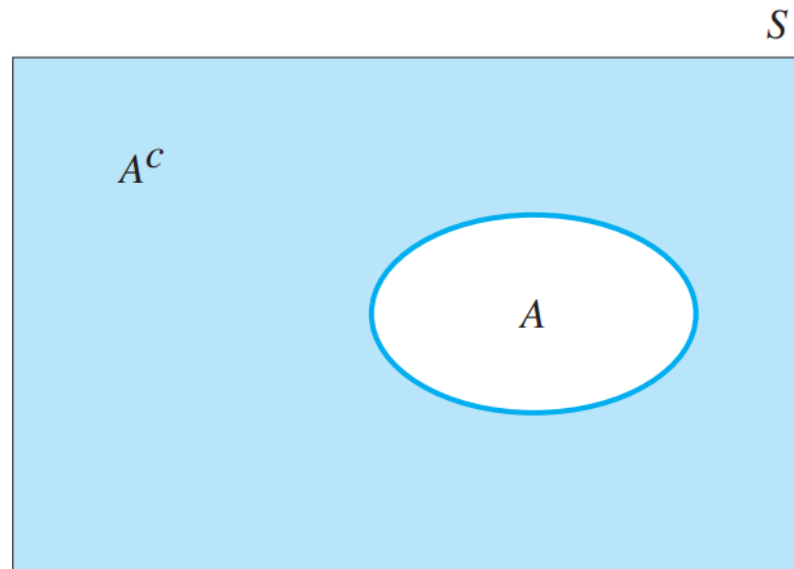
Intersection Rule

The **intersection** of events A and B , denoted by $A \cap B$, is the event that both A and B occur.



Complement Rule

The complement of an event A , denoted by A^c (or A'), is the event that A does not occur





Example

Two fair coins are tossed, and the outcome is recorded. These are the events of interest:

A: Observe at least one head

B: Observe at least one tail

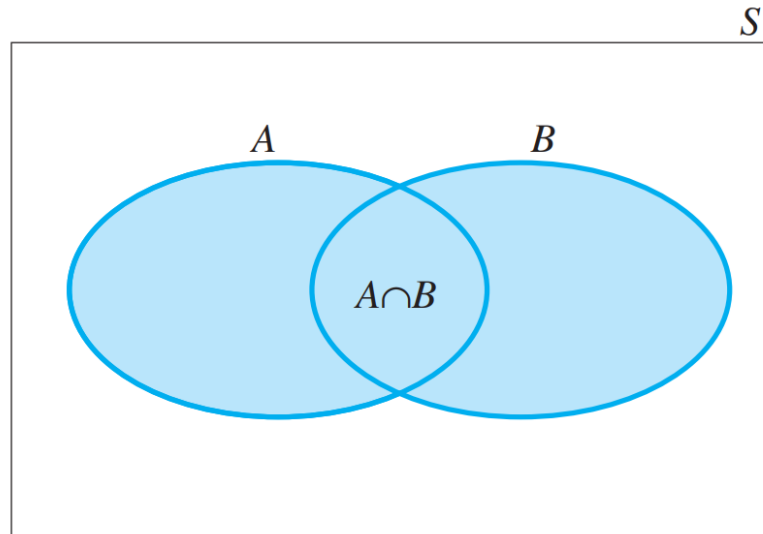
Define the events **A**, **B**, **$A \cup B$** , **$A \cap B$** , and **A^c** as collections of simple events, and find their probabilities

Calculating Probabilities for Unions and Complements

The Addition Rule

Given two events, A and B, the probability of their union, $A \cup B$, is equal to

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$





Mutually Exclusive

What happens if two events are mutually exclusive?



Rule for Complement

$$P(A) + P(A^c) = 1$$

Or

$$P(A) = 1 - P(A^c)$$

Example

An oil-prospecting firm plans to drill two exploratory wells. Past evidence is used to assess the possible outcomes listed in Table

Outcomes for Oil-Drilling Experiment

Event	Description	Probability
A	Neither well produces oil or gas	.80
B	Exactly one well produces oil or gas	.18
C	Both wells produce oil or gas	.02

Find $P(A \cup B)$ and $P(B \cup C)$



The event $A \cup B$ can be described as the event that *at most* one well produces oil or gas, and $B \cup C$ describes the event that *at least* one well produces gas or oil.

Independence, Conditional Probability and The Multiplication Rule



Independent and Dependent Events

There is a probability rule that can be used to calculate the probability of the intersection of **several events**. However, this rule depends on the important statistical concept of independent or dependent events.



Definition

Two events, A and B, are said to be **independent** if and only if the probability of event B is not influenced or changed by the occurrence of event A, or vice versa.



Given

The probability of an event A, given that the event B has occurred, is called the conditional probability of A, given that B has occurred, denoted by $P(A|B)$



General Multiplicative Rule

The probability that both A and B occur when the experiment is performed is

$$P(A \cap B) = P(A) * P(B|A)$$



Conditional Probability

$$P(A|B) = P(A \cap B) / P(B) \text{ Given } P(B) \neq 0$$

$$P(B|A) = P(A \cap B) / P(A) \text{ Given } P(A) \neq 0$$